In [ ]:

```
                                                    Practical No : 05

Name : Dhanke Anuj Mahadev
Roll No : 20
Batch : A1
Problem Statement : ImplementK-Meansclustering/hierarchicalclusteringon
sales_data_sample.csv dataset. Determine the number of clusters using the
elbow method.
```

In [1]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster
```

In [2]:

```
df = pd.read_csv("sales_data_sample.csv", encoding="latin1")
print("Dataset loaded successfully")
print(df.head())
```

```
Dataset loaded successfully
   ORDERNUMBER  QUANTITYORDERED  PRICEEACH  ORDERLINENUMBER    SALES  \
0        10107               30      95.70                2  2871.00
1        10121               34      81.35                5  2765.90
2        10134               41      94.74                2  3884.34
3        10145               45      83.26                6  3746.70
4        10159               49     100.00               14  5205.27

         ORDERDATE   STATUS  QTR_ID  MONTH_ID  YEAR_ID  ...  \
0   2/24/2003 0:00  Shipped       1         2     2003  ...
1    5/7/2003 0:00  Shipped       2         5     2003  ...
2    7/1/2003 0:00  Shipped       3         7     2003  ...
3   8/25/2003 0:00  Shipped       3         8     2003  ...
4  10/10/2003 0:00  Shipped       4        10     2003  ...

                    ADDRESSLINE1  ADDRESSLINE2           CITY STATE  \
0         897 Long Airport Avenue           NaN            NYC    NY
1              59 rue de l'Abbaye           NaN         Reims   NaN
2  27 rue du Colonel Pierre Avia           NaN         Paris   NaN
3              78934 Hillside Dr.           NaN       Pasadena    CA
4                 7734 Strong St.           NaN  San Francisco    CA

  POSTALCODE COUNTRY TERRITORY CONTACTLASTNAME CONTACTFIRSTNAME DEALSIZE
0      10022     USA       NaN              Yu              Kwai    Small
1      51100  France      EMEA         Henriot              Paul    Small
2      75508  France      EMEA        Da Cunha            Daniel   Medium
3      90003     USA       NaN           Young             Julie   Medium
4        NaN     USA       NaN           Brown             Julie   Medium

[5 rows x 25 columns]
```

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

In [3]:

```
num_df = df.select_dtypes(include=[np.number])
```

In [4]:

```
num_df = num_df.dropna()
```

In [5]:

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(num_df)
```

In [6]:

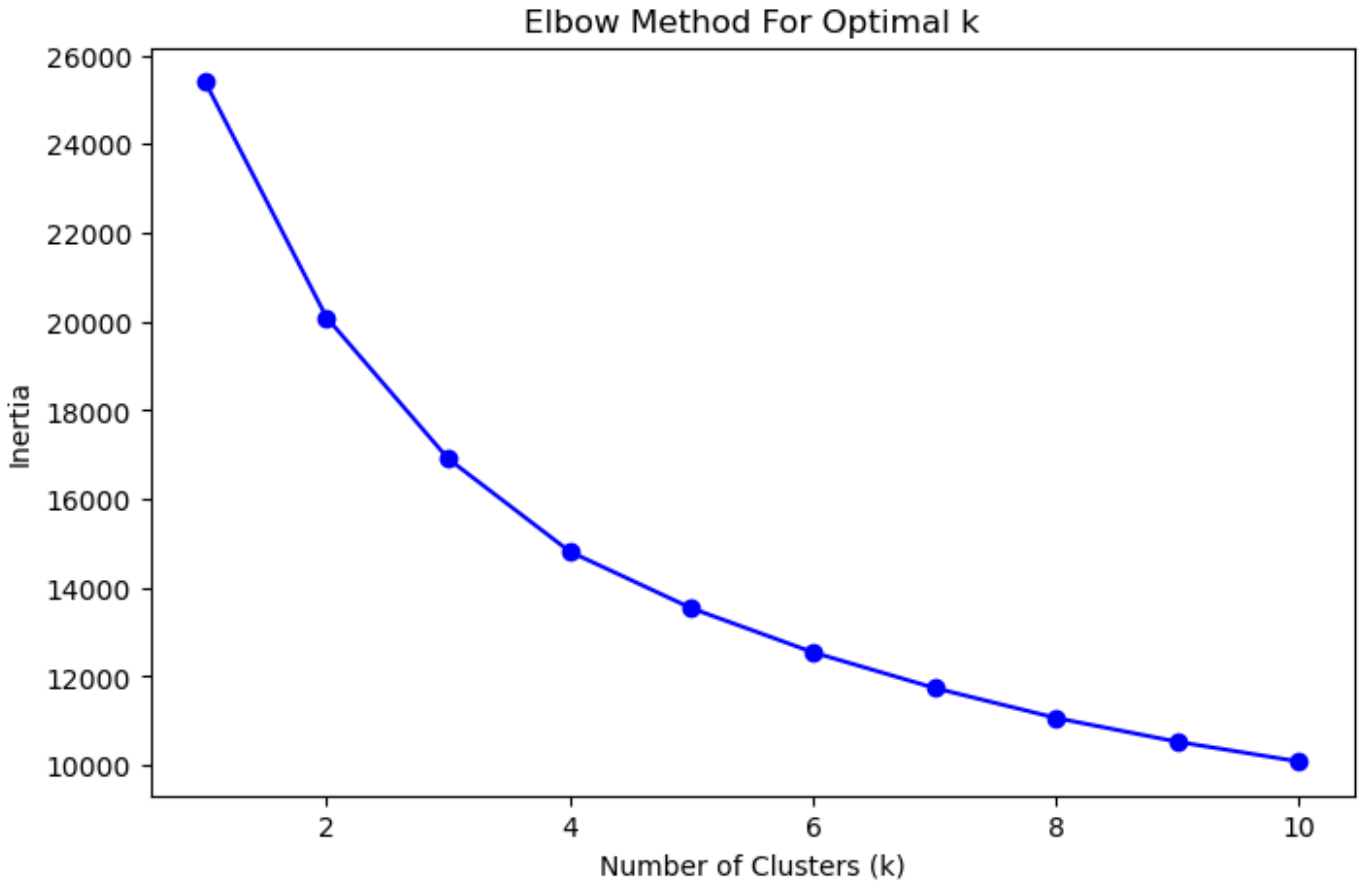
```
inertia = []
K = range(1, 11)
```

In [8]:

```
for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)
```

In [9]:

```
plt.figure(figsize=(8,5))
plt.plot(K, inertia, 'bo-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method For Optimal k')
plt.show()
```



In [10]:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
clusters = kmeans.fit_predict(X_scaled)
```

In [11]:

```
df['KMeans_Cluster'] = clusters
print("\nKMeans cluster counts : ")
print(df['KMeans_Cluster'].value_counts())
```

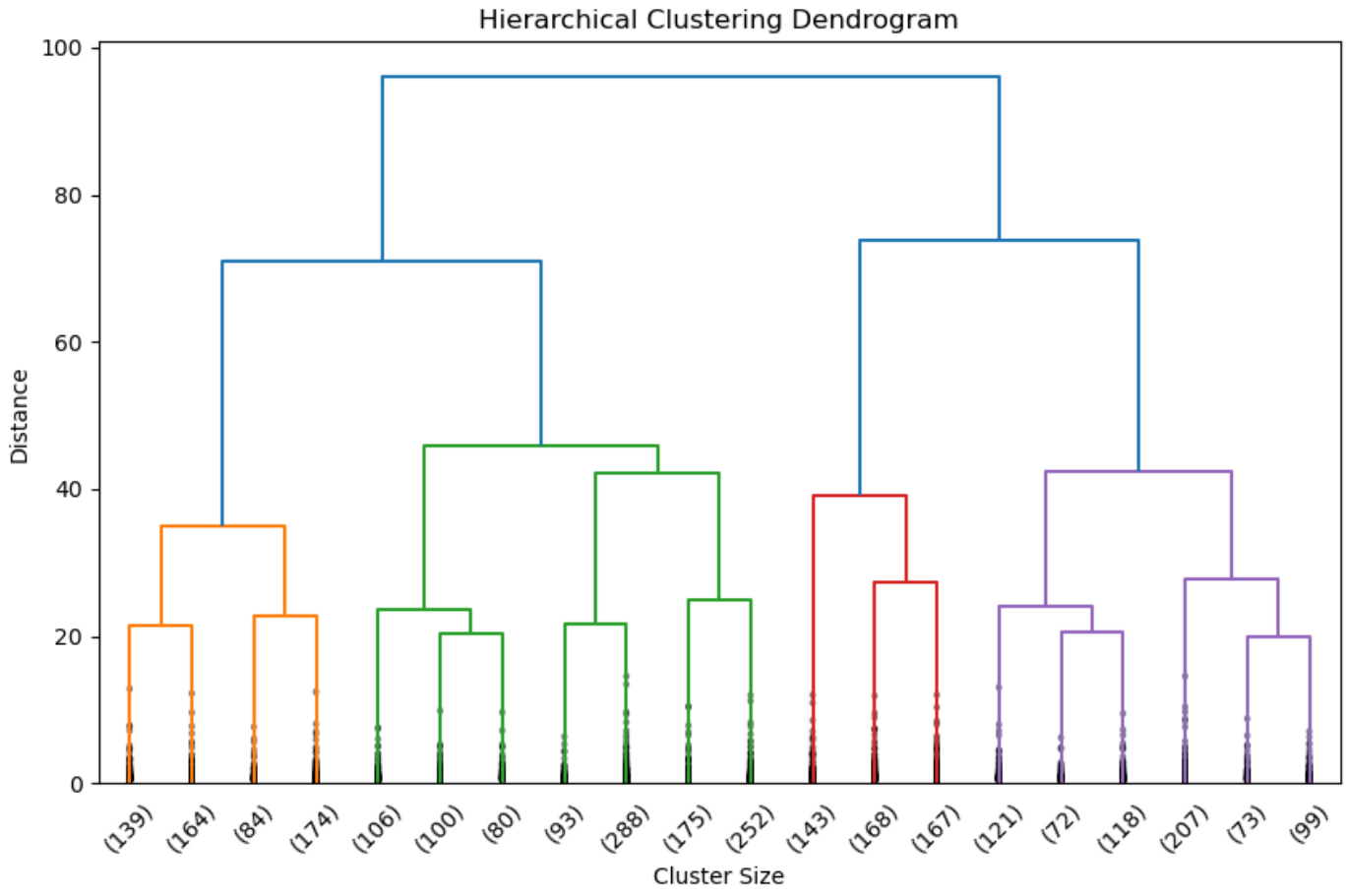
```
KMeans cluster counts :
KMeans_Cluster
1    1183
2     906
0     734
Name: count, dtype: int64
```

In [12]:

```
linked = linkage(X_scaled, method='ward')
```

In [13]:

```
plt.figure(figsize=(10,6))
dendrogram(linked, truncate_mode='lastp', p=20,
           leaf_rotation=45, leaf_font_size=10, show_contracted=True)
plt.title('Hierarchical Clustering Dendrogram')
plt.xlabel('Cluster Size')
plt.ylabel('Distance')
plt.show()
```



In [14]:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
hc_clusters = fcluster(linked, optimal_k, criterion='maxclust')
df['Hierarchical_Cluster'] = hc_clusters
```

In [15]:

```
print("\nHierarchical cluster counts : ")
print(df['Hierarchical_Cluster'].value_counts())
```

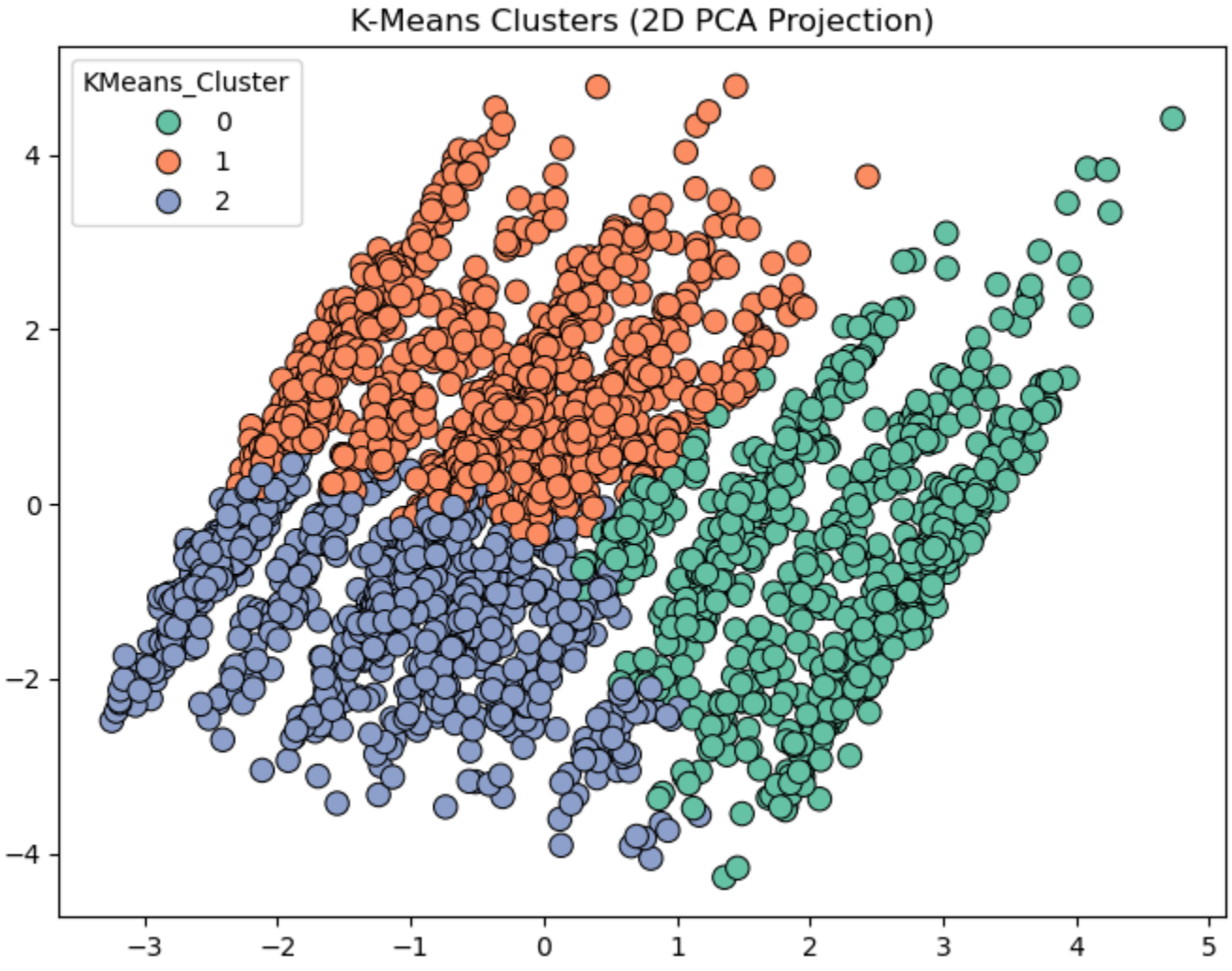
```
Hierarchical cluster counts :
Hierarchical_Cluster
1    1655
3     690
2     478
Name: count, dtype: int64
```

In [16]:

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
```

In [17]:

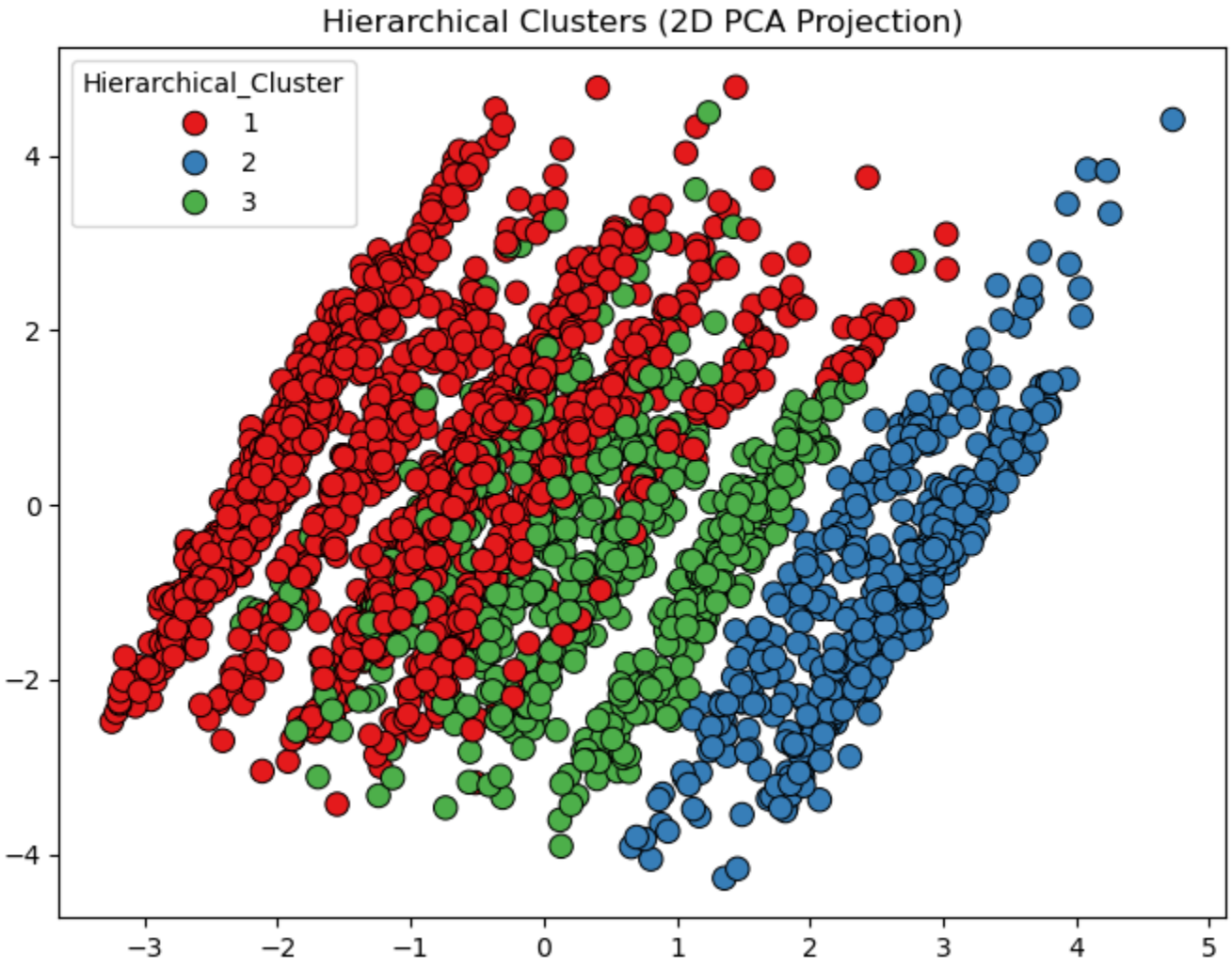
```
plt.figure(figsize=(8,6))
sns.scatterplot(x=X_pca[:,0], y=X_pca[:,1], hue=df['KMeans_Cluster'],
                palette="Set2", s=80, edgecolor="k")
plt.title("K-Means Clusters (2D PCA Projection)")
plt.show()
```



In [18]:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

```
plt.figure(figsize=(8,6))
sns.scatterplot(x=X_pca[:,0], y=X_pca[:,1], hue=df['Hierarchical_Cluster'],
                palette="Set1", s=80, edgecolor="k")
plt.title("Hierarchical Clusters (2D PCA Projection)")
plt.show()
```



In [ ]:

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF